

Supply Chain Management Analysis Documentation

Comprehensive Project Report

Inventory · Demand · Supplier · Warehouse · Risk · Optimization

Prepared for: Interviewers & Clients

Dataset: Inventory_Data.csv | Language: Python 3

Tools: pandas · numpy · matplotlib · seaborn

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Analysis Modules

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Derived Features

5

Key Business Outcomes

This document explains every analytical step performed in the Jupyter Notebook — covering methodology, business rationale, and key insights — so stakeholders can fully understand and act on the results.

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1. Project Overview & Objectives

This project performs a **comprehensive supply chain analysis** using Python. The dataset contains inventory, supplier, warehouse, and demand-related information captured across multiple SKUs and warehouse locations. The goal is to transform raw operational data into actionable intelligence that drives better supply chain decisions.

Business Objectives

- Identify delivery delays and highlight regions / vendors with poor performance.
- Evaluate vendor efficiency by comparing delay percentages.
- Analyse logistics costs across shipment modes to find optimisation opportunities.
- Discover high-demand products to support inventory planning.
- Provide actionable insights to improve delivery timelines, reduce costs, and enhance performance.

Dataset At a Glance

Column	Description
SKU_ID	Unique product identifier
Category	Product category (Beverages, Dairy, Frozen ...)
Warehouse_ID	Warehouse location code
Supplier_ID / Name	Supplier reference
Quantity_On_Hand	Physical units currently in stock
Avg_Daily_Sales	Average units sold per day
Lead_Time_Days	Days from order placement to receipt
Reorder_Point	Stock level that triggers a new order
Safety_Stock	Buffer stock to absorb demand variability
Total_Inventory_Value_USD	Total monetary value of stock
Supplier_OnTime_Pct	% of orders delivered on time
Expiry_Date	Product expiry for perishables
Forecast_Next_30d	Demand forecast for the next 30 days
Damaged_Qty	Units damaged and not sellable
Returns_Qty	Units returned by customers

2. Methodology — 12-Step Workflow

The project follows a structured, repeatable analytical process. Each step builds on the previous to ensure data quality and analytical rigour.

1

Data Collection

The `Inventory_Data.csv` dataset — containing inventory, supplier, warehouse and demand details — is loaded into a pandas `DataFrame` for analysis.

2

Data Understanding

Initial exploration using `df.head()`, `df.shape`, `df.info()`, and `df.describe()` reveals the structure, data types, column names, and basic statistics of the dataset.

3

Data Cleaning & Preprocessing

Missing values, incorrect formats, duplicate records, and type mismatches are corrected. Date columns are parsed, numeric columns are cast to `int/float`, and currency fields are standardised.

4

Feature Engineering

New derived columns — `Available_Stock`, `Stock_Status`, `Coverage_Gap`, `Inventory_Turnover`, `Demand_Supply_Ratio`, `Supplier_Risk`, `SC_Risk_Score`, `Optimization_Score` — are created to enrich the dataset with business-relevant signals.

5

Exploratory Data Analysis (EDA)

Statistical summaries and visualisations reveal patterns, distributions, outliers, and correlations across key variables.

6

Inventory Analysis

Stock levels are evaluated to classify products as Critical, Low, Healthy, or Excess, and Turnover Categories (Slow / Medium / Fast / Very Fast Moving) are assigned.

7

Demand Analysis

Category-level average daily sales are compared and a Demand–Supply Ratio is computed to flag High / Medium / Low risk SKUs.

8

Supplier Performance Analysis

A composite `Supplier_Risk` score combining lead time and on-time delivery rate classifies suppliers into A-Good, B-Normal, and C-High Risk tiers.

9

Warehouse Analysis

Total and average stock levels per warehouse are computed; stock distribution and imbalance are visualised using bar charts and box plots.

10

Risk Analysis

Stockout_Risk flags SKUs where inventory days are less than lead time; Coverage_Gap quantifies the severity; Risk_Level (High / Medium / Low) is assigned per warehouse.

11

Reorder & Optimisation Analysis

Reorder points are recalculated as $\text{Avg_Daily_Sales} \times \text{Lead_Time_Days}$; an Optimization_Score combines Coverage_Gap, Stockout_Risk, and Reorder_Flag to prioritise items needing immediate action.

12

Visualisation & Reporting

Key insights are communicated through bar charts, box plots, histograms, and cross-tabulation heatmaps built with matplotlib and seaborn.

3. Data Collection & Understanding

3.1 Loading the Dataset

The dataset was collected in CSV format and loaded using `pd.read_csv()`. This gives us a pandas DataFrame — a powerful, spreadsheet-like data structure that supports filtering, grouping, and computation.

3.2 Initial Exploration

- `df.head()` — Displays the first 5 rows to visually confirm the data was loaded correctly.
- `df.shape` — Returns (rows, columns), giving an immediate sense of dataset size.
- `df.info()` — Lists every column with its data type and count of non-null values, essential for spotting type problems and missing data.
- `df.columns` — Lists all column names for reference.
- `df.describe()` — Provides count, mean, std, min, quartiles, and max for every numeric column — a quick statistical health-check.

Why This Matters

Understanding the shape and structure of the data before any analysis prevents incorrect assumptions. For example, columns stored as 'object' instead of 'float' would silently fail in mathematical operations — catching this early saves hours of debugging later.

4. Data Cleaning & Preprocessing

4.1 Missing Values

`df.isnull().sum()` counts null entries per column. Missing values cause computation errors and bias results, so they must be handled before analysis.

Column	Strategy	Reason
Stock_Age_Days	Mean per Warehouse_Location group	Location-level mean is a realistic proxy
Damaged_Qty	Mean per Warehouse_Location, then 0	Damaged qty is 0 when not recorded
Avg_Daily_Sales	Median per Category group	Median is robust to outliers in sales data
Notes	Filled with 'No Notes'	Text column — a default string keeps it usable

4.2 Duplicate Records

`df.duplicated().sum()` checks for exact row duplicates. Duplicates inflate counts, distort averages, and introduce bias. Any found would be removed with `df.drop_duplicates()`.

4.3 Date Parsing

Columns *Received_Date*, *Last_Purchase_Date*, *Expiry_Date*, and *Audit_Date* are converted to Python datetime objects using `pd.to_datetime(dayfirst=True)`. This enables date arithmetic — such as calculating stock age or days until expiry.

4.4 Data Type Corrections

Several columns were stored as strings with formatting characters (commas as decimal separators, dollar signs, etc.) that prevented numeric operations. Each column was cleaned and cast to the correct type:

- **Integer columns** (*Stock_Age_Days*, *Quantity_On_Hand*, *Damaged_Qty*, etc.) — stripped of non-numeric characters, converted with `pd.to_numeric()`, NaNs filled with 0.
- **Float columns** (*Total_Inventory_Value_USD*, *Last_Purchase_Price_USD*, *Unit_Cost_USD*) — dollar signs and thousand separators removed, decimal commas replaced, formatted as currency strings for display.
- **Forecast_Next_30d** — dots and commas removed to extract clean integers.
- **SKU_Churn_Rate**, **Avg_Daily_Sales**, **Days_of_Inventory**, **Order_Frequency_per_month**, **Supplier_OnTime_Pct**, **Demand_Forecast_Accuracy_Pct** — comma-to-dot conversion applied and cast to float.

Business Rationale

Correct data types are the foundation of all downstream analysis. An incorrect type means Python performs string concatenation instead of addition, producing completely wrong results that are hard to detect. Every type conversion here directly enables the KPIs and charts in later sections.

5. Feature Engineering

Feature engineering creates new, analytically meaningful columns from existing data. These derived features encode business logic and make patterns machine-readable.

Available_Stock

Formula: $\text{Quantity_On_Hand} - \text{Quantity_Reserved} - \text{Quantity_Committed} - \text{Damaged_Qty}$

Represents the actual sellable inventory. Stock on hand includes reserved and committed units that are already allocated — subtracting them gives the true available quantity for fulfilling new orders.

New_Stock_Status

Rule: $\text{Available_Stock} \leq 0 \rightarrow \text{Out of Stock} \mid < 50 \rightarrow \text{Low} \mid \text{else} \rightarrow \text{In Stock}$

A simple operational label used on warehouse dashboards. Immediately tells operations teams which SKUs need urgent attention without reading raw numbers.

Days_of_Inventory

Formula: $\text{Quantity_On_Hand} \div \text{Avg_Daily_Sales}$

How many days the current stock will last at the current selling rate. This is the core metric for inventory planning and directly feeds the Stock_Status and Coverage_Gap calculations.

Stock_Status

Rule: $\leq 10 \text{ days} \rightarrow \text{Critical} \mid 10-30 \rightarrow \text{Low} \mid 30-60 \rightarrow \text{Healthy} \mid > 60 \rightarrow \text{Excess}$

Classifies inventory health in business terms. Critical SKUs need emergency orders; Excess SKUs tie up capital. This four-tier classification drives replenishment prioritisation.

Coverage_Gap

Formula: $\max(0, \text{Lead_Time_Days} - \text{Days_of_Inventory})$

Positive value means the stock will run out before the next shipment arrives. This quantifies the severity of the stockout risk in days, enabling prioritisation of the most urgent replenishments.

Rate_Level

pd.cut on SKU_Churn_Rate: $0-3 \rightarrow \text{Low} \mid 3-10 \rightarrow \text{Medium} \mid 10-25 \rightarrow \text{High}$

Categorises how fast SKUs are churning (being replaced or discontinued). High-churn products require more frequent ordering cycles.

Inventory_Turnover

Formula: $\text{Avg_Daily_Sales} \times 365 \div \text{Quantity_On_Hand}$

Annual turnover rate. A high number means fast-moving stock (good liquidity); a low number indicates slow-moving stock (capital tie-up risk). Labelled as Very Fast / Fast / Medium / Slow Moving.

Demand_Supply_Ratio

Formula: $\text{Avg_Daily_Sales} \div \text{Quantity_On_Hand}$

Directly measures whether supply keeps pace with demand. Ratio $\geq 0.2 \rightarrow$ High Risk (stock likely insufficient); 0.1–0.2 $\rightarrow$ Medium Risk; $< 0.1 \rightarrow$ Low Risk.

Supplier_Risk

Formula: $\text{Lead_Time_Days} \times (1 - \text{Supplier_OnTime_Pct})$

Combines how long a supplier takes AND how unreliable they are. A supplier with 20-day lead time and 70% on-time rate scores 6 — higher than a 10-day, 100% supplier who scores 0. Classified as A-Good / B-Normal / C-High Risk.

SC_Risk_Score

Formula: $\text{Supplier_Risk} + \text{Coverage_Gap} (\text{Inventory_Risk})$

An aggregate supply chain vulnerability score that combines supplier unreliability and inventory shortage simultaneously. Higher score = higher disruption probability.

Optimization_Score

Formula: $\text{Coverage_Gap} \times 2 + \text{Stockout_Risk} \times 5 + \text{Reorder_Flag} \times 3$

A composite prioritisation score for operations. Weights reflect business impact: stockout risk is weighted highest ($\times 5$) because a stockout means lost sales; coverage gap is next ($\times 2$); reorder flag is the baseline trigger ($\times 3$).

Reorder_Point (recalculated)

Formula: $\text{Avg_Daily_Sales} \times \text{Lead_Time_Days}$

The stock level at which a new order must be placed immediately to avoid a stockout during the supplier's lead time. Compared against Quantity_On_Hand to set the Reorder_Flag.

6. Inventory Analysis

Inventory analysis answers: *Do we have the right products, in the right quantities, at the right locations?* It identifies over-stocked, under-stocked, and slow-moving items, enabling data-driven replenishment and capital optimisation.

6.1 Stock Availability Status

After computing **Available_Stock**, each SKU is labelled as **Out of Stock**, **Low**, or **In Stock**. A countplot visualises the distribution so operations managers can instantly see how many SKUs are in each state.

Chart Interpretation

If a large proportion of bars fall in 'Low' or 'Out of Stock', it indicates systemic under-stocking — likely requiring a review of safety stock policies or supplier lead times.

6.2 Stock Status (Days of Inventory Classification)

Using **Days_of_Inventory** (on-hand ÷ daily sales), each SKU is classified:

Status	Days of Inventory	Implication	Action
Critical	≤ 10 days	Stockout imminent	Emergency replenishment
Low	10–30 days	Below safe buffer	Expedite orders
Healthy	30–60 days	Optimal range	Maintain current policy
Excess	> 60 days	Capital tied up	Reduce future orders / promotions

6.3 Inventory Coverage Gap

The Coverage Gap = $\max(0, \text{Lead_Time_Days} - \text{Days_of_Inventory})$. A positive gap means a SKU will run out of stock **before** the next shipment arrives. A histogram with KDE curve shows the distribution of gap sizes across all SKUs — allowing the team to see how widespread and severe the stockout risk is.

6.4 Inventory Turnover & Movement Categories

Turnover = $\text{Avg_Daily_Sales} \times 365 \div \text{Quantity_On_Hand}$. Products are classified:

- **Very Fast Moving** ($\geq 50\times/\text{year}$) — high demand, low stock tolerance, frequent replenishment needed.
- **Fast Moving** ($20\text{--}50\times/\text{year}$) — healthy turnover, monitor for stockouts.
- **Medium Moving** ($5\text{--}20\times/\text{year}$) — balanced; standard replenishment cycles.
- **Slow Moving** ($< 5\times/\text{year}$) — risk of obsolescence and excess capital lock-in.

6.5 Cross-Analysis: Stock Level vs Movement Type

A grouped bar chart and heatmap cross-tabulate **Stock_Status** against **Turnover_Category**. This is the most powerful inventory insight: **Fast-Moving products in Critical/Low stock** are the most dangerous

combination — high demand and low supply means imminent revenue loss. The heatmap cell colours make these dangerous combinations immediately visible.

Key Finding

A high concentration of fast-moving products in Critical and Low stock levels reveals a severe demand-supply imbalance. Without corrective action, this leads to stockouts, lost revenue, and customer dissatisfaction.

7. Demand Analysis

Demand analysis answers: *Which products are selling fastest, and is our inventory aligned with that demand?* Without understanding demand patterns, replenishment decisions are made on guesswork.

7.1 Average Daily Sales by Category

`df.groupby('Category')['Avg_Daily_Sales'].mean()` computes the average daily units sold per product category, revealing which categories drive the most volume.

Category	Demand Level	Observation
Beverages	Highest (~53 units/day)	Fast-moving; frequent replenishment essential
Fresh Produce	High (~39 units/day)	Perishable; daily monitoring required
Bakery	High (~39 units/day)	Short shelf-life; tight inventory management
Dairy	Moderate	Balance stock to avoid spoilage
Frozen	Moderate	Lower urgency but monitor safety stock
Household	Lower	Slow-moving; risk of overstock
Personal Care	Lower	Slow-moving; reduce order frequency
Pantry	Lowest	Careful management to reduce holding costs

7.2 Demand–Supply Ratio & Risk Levels

The Demand–Supply Ratio = `Avg_Daily_Sales ÷ Quantity_On_Hand` measures whether demand is outpacing supply on a per-SKU basis. Unlike looking at stock levels alone, this ratio contextualises stock against actual sales velocity.

Risk Level	Ratio Range	Meaning	Recommended Action
High Risk	≥ 0.2	Sales very high vs stock	Urgent reorder
Medium Risk	0.1–0.2	Monitor closely	Plan replenishment soon
Low Risk	< 0.1	Stock exceeds demand	Hold or reduce future orders

Key Finding

The majority of SKUs fall in Low Risk, indicating adequate or excess supply. However, the High Risk SKUs — though fewer in number — represent items where stockouts are imminent and demand is strong. These require priority procurement action.

8. Supplier Performance Analysis

Supplier performance analysis evaluates how reliably and quickly suppliers deliver goods. Poor supplier performance is one of the biggest drivers of stockouts and supply chain disruptions.

8.1 Supplier Risk Score

Supplier_Risk = Lead_Time_Days × (1 – Supplier_OnTime_Pct)

This composite metric penalises both long lead times AND poor reliability. A supplier who is slow but always on time scores lower than a supplier who is faster but frequently late. This reflects real supply chain risk more accurately than measuring lead time or on-time rate in isolation.

Scenario	Lead Time	On-Time %	Supplier_Risk Score	Class
Best case	7 days	100%	0.0	A – Good
Typical	10 days	85%	1.5	A – Good
Moderate risk	12 days	75%	3.0	B – Normal
High risk	15 days	65%	5.25	C – High Risk
Worst case	20 days	60%	8.0	C – High Risk

8.2 Supplier Classification (ABC)

Suppliers are grouped into three tiers based on their risk score:

- **A – Good (score ≤ 2):** Reliable suppliers. Maintain current relationships.
- **B – Normal (score 2–5):** Acceptable but monitor; request improvement plans if score rises.
- **C – High Risk (score > 5):** Unreliable suppliers. Consider dual-sourcing or replacement.

8.3 Top High-Risk Suppliers

A bar chart of the top 10 suppliers by average Supplier_Risk score identifies the most problematic vendors (e.g., S001, S004). These suppliers have the greatest negative impact on supply chain continuity and should be flagged for vendor reviews.

8.4 Total Supply Chain Risk Score

SC_Risk_Score = Supplier_Risk + Coverage_Gap — combines external risk (supplier unreliability) with internal risk (insufficient inventory). A high score means the item is at risk from both ends: the supplier may be late AND the inventory buffer is insufficient to absorb the delay.

Key Finding

Most suppliers fall into Class A (Good), indicating the supplier base is broadly reliable. The small number of Class C suppliers, however, represent concentrated risk — a single late delivery from them can cause cascading stockouts across dependent SKUs.

9. Warehouse Analysis

Warehouse analysis examines how inventory is distributed across facilities. Uneven distribution — where some warehouses are overstocked and others understocked — signals inefficiency, unnecessary holding costs, and missed service opportunities.

9.1 Total & Average Stock per Warehouse

Grouping by Warehouse_ID and summing Quantity_On_Hand reveals absolute stock levels. Averaging instead shows typical SKU stock per warehouse (controlling for the number of SKUs stored). Bar charts make disparities between warehouses visually obvious.

9.2 Stock Distribution & Imbalance

A **box plot** of Quantity_On_Hand per Warehouse_ID reveals the spread and outliers. Wide boxes and long whiskers indicate high variability — some SKUs are heavily overstocked while others are critically short in the same warehouse.

The **Warehouse_Stock_Imbalance** column (standard deviation of Quantity_On_Hand within each warehouse group) quantifies this dispersion:

Imbalance Level	Std Dev (~)	Interpretation
Moderate Imbalance	≤ 195	Stock is relatively consistent across SKUs
High Variation	195–210	Noticeable spread; some SKUs need attention
Very Uneven	> 210	Significant imbalance; redistribution recommended

9.3 Stock by Category

Grouping by Category and summing Quantity_On_Hand identifies which product types dominate warehouse space. Categories with excess stock may need order reduction; categories with low stock across all warehouses need restocking.

9.4 Average Lead Time by Supplier

A bar chart of mean Lead_Time_Days per Supplier_Name reveals which suppliers consistently take longer. Warehouses supplied by slow vendors need larger safety stock buffers to compensate — this links warehouse and supplier strategy directly.

10. Reorder & Risk Analysis

Reorder and risk analysis ensures the right products are re-ordered at the right time to prevent stockouts without over-ordering. It translates data signals into concrete procurement triggers.

10.1 Reorder Point Calculation

Reorder_Point = Avg_Daily_Sales × Lead_Time_Days

This formula defines the stock level below which an order must be placed immediately to avoid running out during the supplier's lead time. When `Quantity_On_Hand` falls below `Reorder_Point`, the **Reorder_Flag** is set to `True`.

Worked Example

Product A: `Avg_Daily_Sales` = 20 units, `Lead_Time` = 7 days → `Reorder_Point` = 140 units. If the warehouse has 130 units on hand, the `Reorder_Flag` fires — an order must be placed today.

10.2 Stockout Risk Indicator

Stockout_Risk = Days_of_Inventory < Lead_Time_Days (True/False)

A binary flag that is `True` whenever current stock will run out before the next shipment can arrive. This is a critical operational alert. Combined with `Coverage_Gap` (which measures *how many days* the stockout will last), it enables prioritisation.

10.3 Risk Level by Coverage Gap

`Coverage_Gap` is further binned into three risk levels for warehouse-level reporting:

- **High Risk** (`Gap ≥ 5` days): Stockout will last 5+ days — critical.
- **Medium Risk** (`Gap 2–5` days): Manageable if action is taken immediately.
- **Low Risk** (`Gap < 2` days): Minimal exposure; monitor only.

A **stacked bar chart** by `Warehouse_ID` shows how High / Medium / Low risk SKUs are distributed across warehouses, enabling warehouse managers to focus their replenishment efforts where they are most needed.

10.4 Reorder Items per Warehouse

Summing `Reorder_Flag` by `Warehouse_ID` shows how many SKUs in each warehouse currently need to be re-ordered. Warehouses with many flagged SKUs need procurement attention first.

11. Optimization Analysis

Optimisation analysis combines all risk signals into a single prioritisation framework, enabling operations teams to focus on the items and warehouses that will have the greatest positive impact if addressed.

11.1 Optimization Score

$$\text{Optimization_Score} = \text{Coverage_Gap} \times 2 + \text{Stockout_Risk} \times 5 + \text{Reorder_Flag} \times 3$$

Component	Weight	Business Justification
Coverage_Gap	×2	More gap days = more severe stockout duration
Stockout_Risk	×5	Binary flag — highest weight because it signals imminent revenue loss
Reorder_Flag	×3	Order is overdue — medium urgency, needs scheduling

11.2 Top 10 Items Needing Optimization

SKUs are sorted by Optimization_Score descending and the top 10 are visualised in a horizontal bar chart. This is the **daily action list** for inventory managers — the items at the top of this list should receive same-day attention.

11.3 Warehouse Optimization Need

Averaging Optimization_Score by Warehouse_ID shows which warehouse as a whole has the greatest inventory management challenges. A high warehouse-level score might indicate structural problems: poor supplier contracts, inadequate safety stock policies, or SKU rationalisation needs.

Strategic Recommendation

Warehouses with consistently high optimization scores should be prioritised for supply chain process reviews, including supplier negotiations, safety stock recalibration, and demand forecasting improvements.

12. Key Outcomes & Business Recommendations

Outcome 1: Delivery Delay Identification

Finding: Analysed the difference between shipment and delivery dates to flag regions and vendors with consistently poor delivery performance.

Recommendation: Hold quarterly vendor performance reviews. Introduce SLA penalties for suppliers with on-time delivery below 80%.

Outcome 2: Vendor Efficiency Evaluation

Finding: Compared supplier delay percentages and classified suppliers into A / B / C tiers based on a composite risk score.

Recommendation: Dual-source from Class C suppliers. Reward Class A suppliers with preferred vendor status to incentivise continued reliability.

Outcome 3: Logistics Cost Analysis

Finding: Identified variations in logistics costs across shipment modes (ground, air, sea), enabling cost-versus-speed trade-off analysis.

Recommendation: Shift low-urgency, slow-moving SKUs to sea freight. Reserve air freight for High Risk / Critical stock items only.

Outcome 4: High-Demand Product Discovery

Finding: Beverages and Fresh Produce consistently show the highest average daily sales, placing them at the top of replenishment priority.

Recommendation: Set tighter safety stock thresholds for these categories. Increase order frequency and consider JIT (just-in-time) delivery contracts.

Outcome 5: Inventory Risk Mitigation

Finding: Fast-moving products in Critical and Low stock represent the most dangerous combination — high demand with insufficient supply.

Recommendation: Implement automated reorder triggers for all Fast and Very Fast Moving SKUs currently in Critical or Low stock status.

This documentation was produced to provide complete transparency into the analytical methodology, business logic, and insights generated from the Supply Chain Management Analysis project. Each analysis module builds on the previous to deliver an integrated view of inventory health, demand alignment, supplier

reliability, and operational risk.